Optimization

Foundations of Mathematics 11 — BC Curriculum

Today's Lesson at a Glance

What we will cover today:

1. What is optimization?
2. Constraints & feasible regions
3. The corner-point method
4. Geometric optimization

Tools: **Desmos**, pencil & paper, partner discussions

Learning Goals

By the end of class you will be able to:

- Write real-world constraints as inequalities
- Identify the feasible region on a graph
- Use corner points to find the best possible value
- Solve profit and area optimization problems

Part 1

What Is Optimization?

$\approx$ 10 minutes

What Does It Mean to "Optimize"?

To **optimize** means to find the best possible value given the limits we are working with.

- **Maximize** — make as *large* as possible
  e.g. profit, area, number of items produced
- **Minimize** — make as *small* as possible
  e.g. cost, waste, fencing needed

There is always a catch: real situations have **constraints** (limits) that restrict our choices.

Quick Example

Maya earns \$15/h at her part-time job.

She wants to work 100 h a week.
She can only work 20 h a week.

The **constraint** is 20 h/week.
The **optimal** choice is to work all 20 available hours.

Optimization Is Everywhere

Business

A bakery wants to **maximize profit** selling muffins and loaves, given limited flour, butter, and baking time.

Agriculture

A farmer wants to **maximize the area** of a rectangular garden using only 80 m of fencing.

Everyday Life

You want to **maximize your free time** on the weekend while still finishing assignments and getting enough sleep.

Pair Discussion (2 min)

Think of one real situation — yours or someone else's — where someone had to optimize something. What was the *goal*? What were the *constraints*?

Part 2

Constraints & Feasible Regions

$\approx$ 20 minutes

Writing Constraints as Inequalities

Scenario: Furniture Workshop

A small workshop makes **chairs** (x) and **tables** (y) each day.

- The machines can build **at most 8 chairs** per day: $x \leq 8$
- Workers can finish **at most 6 tables** per day: $y \leq 6$
- They must fill **at least 4 orders combined**: $x + y \geq 4$
- You cannot make a negative number of items: $x \geq 0, \quad y \geq 0$

Writing Constraints as Inequalities

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Key Term: Feasible Region

The **feasible region** is the set of all points (x, y) that satisfy *every* constraint at the same time. Any point inside the feasible region represents a realistic option.

Graphing the Feasible Region

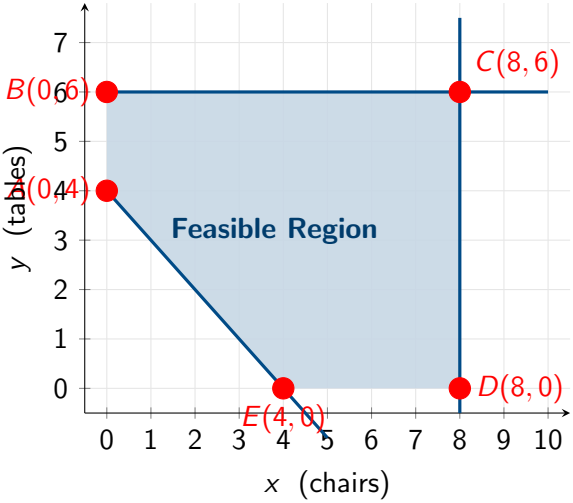
Constraints:

$$x \leq 8$$
$$y \leq 6$$
$$x + y \geq 4$$
$$x \geq 0, \quad y \geq 0$$

Graph each inequality.

The **feasible region** is where all shadings overlap.

The **corner points** are where the boundary lines meet.



[Desmos Activity] — Activity 1: Graph a Feasible Region (10 min)

Instructions

1. Open `desmos.com/calculator` on your device.
2. Type each constraint as an inequality, one per line:

   `x <= 8` `y <= 6` `x + y >= 4` `x >= 0` `y >= 0`

3. Desmos shades each region. The **overlap** (darkest area) is the feasible region.
4. **Click on the boundary lines** to find where they intersect — those are the corner points.

Checkpoint — Compare with a partner

List all five corner points. Write them in your notes.

Answer: $(0,4)$, $(0,6)$, $(8,6)$, $(8,0)$, $(4,0)$

Part 3

The Corner-Point Method

$\approx$ *35 minutes*

The Corner-Point Theorem

Corner-Point Theorem

When optimizing a **linear objective function** over a feasible region, the **maximum** and **minimum** values always occur at one of the **corner points** (vertices) of the feasible region.

This means we only need to check the corners — not every point!

Procedure

1. **Graph** the feasible region; identify all **corner points**.
2. Write the **objective function** (the expression you want to maximize or minimize).
3. **Substitute** each corner point into the objective function.
4. The **largest result** is the maximum; the **smallest** is the minimum.
5. **State your answer in context.**

Worked Example 1: Maximizing Profit

Problem

Using the furniture workshop scenario: profit is \$40 per chair and \$70 per table. What combination **maximizes daily profit**?

Objective function: $P = 40x + 70y$

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Evaluate P at each corner point:

Point	$(x,\ y)$	$P = 40x + 70y$
A	$(0,\ 4)$	$0 + 280 = \$280$
B	$(0,\ 6)$	$0 + 420 = \$420$
C	$(8,\ 6)$	$320 + 420 = \$740$
D	$(8,\ 0)$	$320 + 0 = \$320$

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Answer

Maximum profit $= \$740$

Achieved at $C = (8,\ 6)$:

make **8 chairs** and **6 tables** per day.

Verify $C = (8, 6)$ satisfies all constraints:

Worked Example 2: Catering Company — Set-Up

Problem

A catering company makes **vegetarian meals** (v) and **meat meals** (m) for an event.

Constraints:

- $v \leq 50$ (prep-table capacity)
- $m \leq 80$ (oven capacity)
- $v + m \geq 60$ (minimum order size)
- $v \geq 0, \quad m \geq 0$

Profit:

\$12 per vegetarian meal

\$9 per meat meal

Objective function:

$P = 12v + 9m$

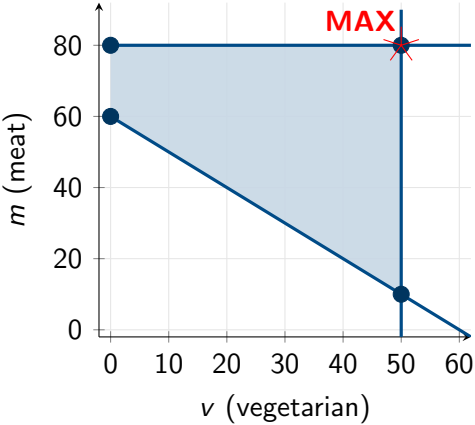
Your Task [Desmos] (5 min)

1. Graph all five constraints in Desmos. Identify the feasible region.
2. Write down all corner point coordinates.
3. Evaluate $P = 12v + 9m$ at each corner. Which gives the maximum?

Worked Example 2: Solution

Corner Points:

$(v,\ m)$	$P = 12v + 9m$
$(0,\ 60)$	$0 + 540 = \$540$
$(0,\ 80)$	$0 + 720 = \$720$
$(50,\ 80)$	$600 + 720 = \$1\,320$
$(50,\ 10)$	$600 + 90 = \$690$



Answer

Maximum profit $= \$1\,320$

Make **50 vegetarian** and **80 meat** meals.

You Try! — Partner Practice

Problem

A student earns money by **mowing lawns** (L) and **walking dogs** (D) each week.

- At most 10 lawn jobs per week: $L \leq 10$
- At most 15 dog walks per week: $D \leq 15$
- At least 8 jobs total per week: $L + D \geq 8$
- $L \geq 0, \quad D \geq 0$

Earnings: \$25 per lawn job, \$15 per dog walk.

Find the combination that **maximizes weekly earnings**.

Steps [Desmos]

1. Write the objective function: $E = \ldots$
2. Graph all constraints in Desmos; identify the corner points.

You Try! — Solution

Objective: $E = 25L + 15D$

Corner Points:

$(L,\ D)$	$E = 25L + 15D$
$(0,\ 8)$	$0 + 120 = \$120$
$(0,\ 15)$	$0 + 225 = \$225$
$(10,\ 15)$	$250 + 225 = \$475$
$(10,\ 0)$	$250 + 0 = \$250$
$(8,\ 0)$	$200 + 0 = \$200$

Answer

Maximum earnings $= \$475$

Do **10 lawn jobs** and **15 dog walks** per week.

Verify constraints at $(10, 15)$**:**

$10 \leq 10$ ✓ $\quad 15 \leq 15$ ✓

$10 + 15 = 25 \geq 8$ ✓

Minimizing Cost — A Different Direction

Problem: Minimizing Cost

A school snack bar makes **sandwiches** (s) and **wraps** (w).

- Must make at least 10 sandwiches: $s \geq 10$
- Must make at least 5 wraps: $w \geq 5$
- Total items cannot exceed 30: $s + w \leq 30$
- $s \geq 0, \quad w \geq 0$

Ingredient cost: \$2 per sandwich, \$3 per wrap. Find the combination that **minimizes total ingredient cost**.

Minimizing Cost — A Different Direction

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- Total items cannot exceed 30: $s + w \leq 30$
- $s \geq 0, \quad w \geq 0$

Ingredient cost: \$2 per sandwich, \$3 per wrap. Find the combination that **minimizes total ingredient cost**.

Solution

Objective: $C = 2s + 3w$

Corner points: $(10, 5)$, $(25, 5)$, $(10, 20)$

$C(10, 5) = 20 + 15 = \$35 \qquad C(25, 5) = 50 + 15 = \$65 \qquad C(10, 20) = 20 + 60 = \$80$

Part 4

Geometric Optimization

$\approx$ 15 minutes

Maximizing Area with Fixed Fencing

Two Classic Questions

- Given a **fixed perimeter**, what rectangle has the **largest area**?
- Given a **fixed area**, what rectangle uses the **least perimeter**?

Setting Up: 40 m of Fencing (all four sides)

Let w = width (m). Then length $\ell = 20 - w$ (since $\ell + w = 20$).

Objective (maximize): $A = \ell \cdot w = (20 - w)\, w$

Let's build a table of values, then graph A vs. w in Desmos to find the maximum.

Exploring with a Table

Perimeter = 40 m $\Rightarrow \ell = 20 - w$

Width w (m)	Length $\ell = 20 - w$ (m)	Area $A = \ell \cdot w$ (m^2)
2	18	36
4	16	64
6	14	84
8	12	96
10	10	100
12	8	96
14	6	84
16	4	64

[Desmos Activity] — Activity 2: Area vs. Width (8 min)

Instructions

1. In Desmos, type: `y = (20 - x) * x`
2. This graphs **Area** (y-axis) vs. **Width** (x-axis).
3. Use the **trace** feature (click on the curve) to slide along the graph and find the highest point.
4. What value of x gives the maximum y?

[Desmos Activity] — Activity 2: Area vs. Width (8 min)

Instructions

1. In Desmos, type: `y = (20 - x) * x`
2. This graphs **Area** (y-axis) vs. **Width** (x-axis).
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What Do You Notice?

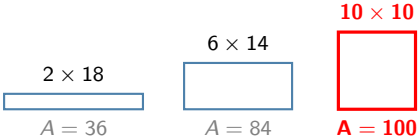
- The graph is a **parabola** (a symmetric curve opening downward).
- The **maximum** is exactly at $x = 10$, $y = 100$.
- Maximum area $= 100\ \text{m}^2$ when $w = 10$ m and $\ell = 10$ m.
- The optimal rectangle is a **square**!

The Square Is Always Optimal

Key Result for Rectangles

- **Fixed perimeter** $\Rightarrow$ the **square** gives the **maximum area**.
- **Fixed area** $\Rightarrow$ the **square** gives the **minimum perimeter**.

All three rectangles below have perimeter $= 40$ m.
Notice how the areas compare:



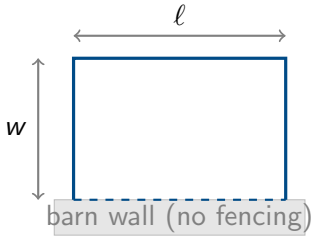
Why?

Equal side lengths mean no "wasted" perimeter. As soon as one side is longer than the other, the area shrinks.

Worked Example: Three-Sided Fencing

Problem

A farmer has **60 m** of fencing for a rectangular pen. One long side is a barn wall — no fencing needed there. Find the dimensions that **maximize the area**.



Constraint:

$\ell + 2w = 60 \;\Rightarrow\; \ell = 60 - 2w$

Objective: $A = \ell \cdot w = (60 - 2w)\, w$

w (m)	$\ell = 60 - 2w$	A (m^2)
5	50	250
10	40	400
15	30	450
20	20	400
25	10	250

Wrap-Up

Summary & Exit Ticket

$\approx$ 5 minutes

Summary: Key Ideas

Linear Optimization

1. Write **constraints** as inequalities.
2. Graph the **feasible region** (use Desmos!).
3. Find all **corner points**.
4. Evaluate the **objective function** at each corner.
5. Pick the **max** or **min** as required.
6. State the answer **in context**.

Geometric Optimization

- Fixed perimeter $\Rightarrow$ **square** maximizes area.
- Fixed area $\Rightarrow$ **square** minimizes perimeter.
- Build a **table** or use **Desmos** to graph and find the vertex.

Always remember

Verify your answer satisfies all constraints.
Interpret your answer in the real-world context.

Vocabulary Recap

Optimize Find the best possible value (max or min).

Constraint A restriction or limit on the variables.

Feasible Region All points satisfying every constraint.

Corner Point A vertex of the feasible region.

Objective Function The expression to maximize or minimize.

Maximum The largest value of the objective function.

Minimum The smallest value of the objective function.

Exit Ticket (5 min)

On a slip of paper:

A store sells **mugs** (m) and **plates** (p). Profit: \$8 per mug, \$5 per plate.

Constraints: $m \leq 20, \;\; p \leq 30, \;\; m + p \geq 15, \;\; m \geq 0, \;\; p \geq 0$

1. Write the **objective function**.
2. List the **corner points** of the feasible region.
3. What combination **maximizes profit**?

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1. Write the **objective function**.
2. List the **corner points** of the feasible region.
3. What combination **maximizes profit**?

Answers

1. $P = 8m + 5p$

2. $(0, 15), \ (0, 30), \ (20, 30), \ (20, 0), \ (15, 0)$

3. $(20, 30)$: $\quad P = 8(20) + 5(30) = 160 + 150 = \$310 \quad$ *(maximum)*

Homework

Practice Problems

1. A manufacturer makes **jackets** (j) and **scarves** (s). Constraints: $j \leq 40$, $s \leq 60$, $j + s \geq 30$, $j, s \geq 0$. Profit: \$35/jacket, \$15/scarf. Find the combination that **maximizes profit**.
2. A rectangular field has a **perimeter of 200 m**. Find the dimensions that **maximize the area**. Confirm your answer using Desmos.
3. **Challenge:** A three-sided rectangular pen uses 100 m of fencing (one side is a wall). Find the dimensions that maximize area. Compare the result to what you found in Question 2.

Great work today!

Desmos is available for all homework problems at `desmos.com/calculator`